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Covariance Estimation Strategies in Massive Spatial and Spatial-Temporal Data

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Abstract:

Covariance plays very important roles in modelling and in prediction of spatial data. For instance, many statistical inferences such as maximum likelihood estimation and best linear unbiased prediction involve covariance matrices which must be inverted and this evaluation is slow when the number of observations is large, and the complexity increases even further in a space-time context. First, we address a Bayesian grouping-based approach, extending Vecchia's approximation and relying on the conditional independence assumption of ordered data which significantly reduces computational burden in nonparametric estimation of a nonstationary spatial covariance. Next, we propose half spectral composite likelihood for the estimation of spatial-temporal covariances which involves a spectral approach in time and a covariance function in space. The approach requires no matrix inversions and the estimators are shown to be consistent and asymptotically normal under increasing domain asymptotic. Examples are given using simulated and real data.

Keywords: Half-spectral model, Composite Likelihood, Whittle likelihood, Vecchia approximation
Mathematics Subject Classification (2010): 60M30, 62M11, 62M30, 62H11.

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1 Introduction

Spatial and Spatial-temporal(ST) data are frequently encountered in many scientific disciplines, especially in environmental, meteorological or geophysical contexts. One major challenge in modeling is the high dimensionality of such data. This is crucial because many statistical inferences such as maximum likelihood estimation and best linear unbiased prediction e.g. kriging, with massive datasets, involve large dimensional covariance matrices which must be inverted. Therefore modeling covariance structures and approximating the likelihood functions are two fundamental keys to tackle such problems.

There have been several computational efficient approaches to overcome the large dimensional matrix problem, such as assuming cyclic stationary process, assuming separability on ST covariance functions, tapering the covariance matrix, including lower-dimensional conditional distributions(Stein et al., 2004), pseudo-likelihood methods, and approximating the random field (RF) with a Gaussian Markov RF(Lindgren et al., 2011). Low-rank approximations such as fixed rank kriging (Cressie and Johannesson, 2008) have also been developed.

The nearest neighbor Gaussian process introduced by Datta et al. (2016) offers a well-defined model for large spatial datasets by constructing sparse precision matrices, enabling efficient Bayesian inference. In this paper, we introduces a Bayesian grouping-based method designed to overcome these limitations and improve modeling of complex spatial structures. Built upon Vecchia’s approximation and accounting for repeated observations, the method enables fast and accurate Bayesian inference (Kidd and Katzfuss, 2022). The core innovation is a spatial grouping algorithm that partitions locations into independent blocks, eliminating weak dependencies and enhancing sparsity in the covariance matrix.

Composite likelihood(CL) as proposed in Lindsay (1988) is also convenient in the setting where the full likelihood is not feasible or is difficult to construct. The idea of CL method is zeroing out the weak correlation between observations at large distances. Varin (2008) provided a survey of CL applications. Earlier, Besag (1975) considered a similar approach to CL for spatial data. Bevilacqua et al. (2012) introduced this approach to a ST framework. Let $Z(\mathbf{s}, t)$ be a real-valued stationary ST process defined on $\mathbb{R}^d \times \mathbb{R}$, where $\mathbf{s} \in \mathbb{R}^d$ represents the spatial site in d dimension, and $t \in \mathbb{R}$ represents the time. Assume that the second moments for the RF exist and are finite. Write the ST process in the general form

$$Z(\mathbf{s}, t) = \int \int e^{i(\mathbf{s}'\boldsymbol{\omega} + t\tau)} d\mathbf{W}(\boldsymbol{\omega}, \tau) = \int e^{it\tau} J(\mathbf{s}, \tau) d\tau, \quad (\mathbf{s}, t) \in \mathbb{R}^d \times \mathbb{R} \quad (1.1)$$

where $J(\mathbf{s}, \tau) = \int_{\omega} e^{i\mathbf{s}'\omega} d\mathbf{W}(\omega, \tau)$, $(\omega, \tau) \in \mathbb{R}^d \times \mathbb{R}$ and $\mathbf{W}$ is a generalized random function with complex symmetry. Note that $d\mathbf{W}(\omega, \tau) = d\mathbf{W}^*(-\omega, -\tau)$ ensures that $Z(\mathbf{s}, t)$ is real-valued. Suppose that C is a continuous function on $\mathbb{R}^d \times \mathbb{R}$. Then C is a covariance function if and only if it is of the form

$$C(\mathbf{h}, u) = \int \int e^{i(\mathbf{h}'\omega + u\tau)} d\mathbf{F}(\omega, \tau) = \int e^{iu\tau} H(\mathbf{h}, \tau) d\tau, \quad (\mathbf{h}, u) \in \mathbb{R}^d \times \mathbb{R} \quad (1.2)$$

where $H(\mathbf{h}, \tau) = \int_{\omega} e^{i\mathbf{h}'\omega} d\mathbf{F}(\omega, \tau)$ and $\mathbf{F}$ is a finite, non-negative and symmetric spectral measure on $\mathbb{R}^d \times \mathbb{R}$. Note that $E|d\mathbf{W}(\omega, \tau)|^2 = d\mathbf{F}(\omega, \tau)$. We shall call H a “half spectral function” since it depends on the spatial lag $\mathbf{h}$ and the temporal frequency τ (Mosammam and Kent, 2016). Here C , f and H can be obtained from each other via the Fourier transforms. Even if these three functions are equivalent, there are situations for which it is more effective to use one domain despite to the other. For example, the half-spectral function H is very useful in practice when we have long records of time series data at a modest number of scattered spatial locations. Notice $H(\mathbf{h}, \tau)$ can be interpreted in two ways. First, write $H(\mathbf{h}, \tau) = k(\tau)\rho(\mathbf{h}, \tau)$ where for a fixed spatial site $\mathbf{s}$, $k(\tau) = H(\mathbf{0}, \tau)$ is the spectral density of the stationary time series $\{Z(\mathbf{s}, t), t \in \mathbb{R}\}$ and for any fixed spatial lag $\mathbf{h}$, $\rho(\mathbf{h}, \tau)$ is the coherence function of the two stationary time series $\{Z(\mathbf{s}, t), t \in \mathbb{R}\}$ and $\{Z(\mathbf{s} + \mathbf{h}, t), t \in \mathbb{R}\}$, each with spectral density $k(\tau)$; for each $\tau \in \mathbb{R}$, $\mathbf{h} \neq \mathbf{0}$, we have $|\rho(\mathbf{h}, \tau)| \leq 1$ and $\rho(\mathbf{0}, \tau) = 1$. Second, for any fixed temporal frequency τ , it is a complex-valued spatial autocorrelation function of the two complex-valued series $\{J(\mathbf{s}, \tau), \mathbf{s} \in \mathbb{R}^d\}$ and $\{J(\mathbf{s} + \mathbf{h}, \tau), \mathbf{s} \in \mathbb{R}^d\}$ as a function of the spatial lag $\mathbf{h}$. Define $\gamma(\mathbf{h}, \tau) = \kappa(\tau) - H(\mathbf{h}, \tau)$ which is generally a complex valued function. By the first interpretation $2\gamma(\mathbf{h}, \tau) = 2\kappa(\tau)(1 - \text{Re}\{\rho(\mathbf{h}, \tau)\})$ is the spectral density of $\{Z(\mathbf{s} + \mathbf{h}, t) - Z(\mathbf{s}, t)\}$ while by the second interpretation $\gamma(\mathbf{h}, \tau)$ is the spatial semi-variogram of $J(\mathbf{s}, \tau)$.

In this paper our second objective is to consider the estimation of the parameters of ST covariance functions using a half-spectral composite likelihood (HSCL) approach which involves a spectral approach in time but a covariance approach in space. The idea of the proposed approach is to facilitate the analysis of spectral density of contrast components at different spatial sites. Then, the Whittle likelihood (WL) approximation is used. The resulting HSCL approach requires no matrix inversions and the estimators are shown to be consistent and asymptotically normal under increasing domain asymptotic in a fashion similar to Bevilacqua et al. (2012).

This paper is organized as follows. In Section 2 Vecchia grouping for Bayesian estimating of nonparametric nonstationary spatial covariances is proposed. HSCL estimation methods for

estimating ST covariances are proposed in Section 3. A simulation study and real data analysis is conducted in Section 4. The paper ends with a conclusion.

2 Vecchia Grouping for Bayesian Estimating Spatial Covariances

Let $\mathbf{Y} = \left\{ \mathbf{y}_i^{(l)} : i = 1, \dots, n, \quad l = 1, \dots, N \right\}$ denote N replicates of a nonstationary Gaussian random field observed at spatial locations $\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_n$. Assume the locations are ordered using the maximum-minimum distance, which selects locations such that the minimum distance between the nearest points in the selected set is maximized. After ordering, let $\mathbf{y}_i = \left(y_i^{(1)}, \dots, y_i^{(N)} \right)'$ denote the observations vector at location $\mathbf{s}_i$, for $i = 1, \dots, n$. Also, define $\mathbf{y}^{(l)} = \left(y_1^{(l)}, \dots, y_n^{(l)} \right)'$ as the spatial observation in the l -th replicate, modeled as $\mathbf{y}^{(l)} \stackrel{iid}{\sim} \mathcal{N}_n(\mathbf{0}, \mathbf{\Sigma})$, $l = 1, \dots, N$, where $\mathbf{\Sigma}$ is the spatial covariance matrix with entries defined by the covariance function $C(\mathbf{s}_i, \mathbf{s}_j)$. The goal is to estimate the spatial covariance matrix $\mathbf{\Sigma}$ or its inverse (the precision matrix) from the observations. Next, we apply Vecchia's approximation

$$p(y_1^{(l)}, \dots, y_n^{(l)} | \mathbf{\Sigma}) = p(y_1^{(l)}) \prod_{i=2}^n p(y_i^{(l)} | y_{j_{i1}}^{(l)}, \dots, y_{j_{im}}^{(l)}, \mathbf{\Sigma}), \quad l = 1, \dots, N \quad (2.1)$$

where $J_m(i) = \{j_{i1}, \dots, j_{im}\}$ is subset of m nearest neighbors among previously ordered locations. This ordered conditional independence structure not only reduces computational complexity for large datasets but also implicitly induces a specific structure in the precision matrix. This structure corresponds to modified Cholesky decomposition, defined as $\mathbf{\Sigma}^{-1} = \mathbf{U}\mathbf{D}^{-1}\mathbf{U}'$ where $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$ is a diagonal matrix with positive entries, and $\mathbf{U}$ is a unit upper triangular matrix. The assumed conditional independence leads to sparsity in $\mathbf{U}$, where each column has at most m nonzero off-diagonal entries (Kidd and Katzfuss, 2022). These nonzero entries in column i are denoted by $\mathbf{u}_i = \mathbf{U}_{J_m(i), i}$, representing the $J_m(i)$ indices in column i of $\mathbf{U}$. Now, we can write as a sequence of n linear regression models as follows:

$$p(\mathbf{Y} | \mathbf{\Sigma}) = \prod_{i=1}^n p(\mathbf{y}_i | \mathbf{y}_{1:i-1}, \mathbf{\Sigma}) = \prod_{i=1}^n \mathcal{N}_N(\mathbf{y}_i | \mathbf{X}_i \mathbf{u}_i, d_i \mathbf{I}_N) \quad (2.2)$$

Here, $\mathbf{I}_N$ is the identity matrix and the design matrix $\mathbf{X}_i$ consists of the observations at the m neighbor locations of $\mathbf{s}_i$, stored in the columns of $\mathbf{Y}$ with indices $J_m(i)$. The Bayesian regression model corresponding to equation (2.2), using independent NIG priors, is defined as follows:

$$\mathbf{u}_i | d_i, \boldsymbol{\theta} \stackrel{\text{ind}}{\sim} \mathcal{N}(0, d_i \mathbf{V}_i), \quad d_i | \boldsymbol{\theta} \stackrel{\text{ind}}{\sim} \mathcal{IG}(\alpha_i, \beta_i), \quad i = 1, \dots, n. \quad (2.3)$$

Here, $\boldsymbol{\theta}$ is the vector of hyperparameters including m , $\mathbf{V}_i$, α_i , and β_i . The joint posterior distribution for the regression coefficients and variances is then computed as:

$$p(\mathbf{u}_1, \dots, \mathbf{u}_n, d_1, \dots, d_n \mid \mathbf{Y}, \boldsymbol{\theta}) \propto \prod_{i=1}^n \mathcal{N}(\mathbf{u}_i \mid \hat{\mathbf{u}}_i, d_i \mathbf{G}_i) \cdot \mathcal{IG}(d_i \mid \tilde{\alpha}_i, \tilde{\beta}_i) \quad (2.4)$$

with corresponding parameters given by:

$$\begin{aligned} \hat{\mathbf{u}}_i &= \mathbf{G}_i \mathbf{X}_i' \mathbf{y}_i, \quad \mathbf{G}_i = (\mathbf{X}_i' \mathbf{X}_i + \mathbf{V}_i^{-1})^{-1}, \quad \tilde{\alpha}_i = \alpha_i + \frac{N}{2}, \\ \tilde{\beta}_i &= \beta_i + \frac{1}{2} (\mathbf{y}_i' \mathbf{y}_i - \hat{\mathbf{u}}_i' \mathbf{G}_i^{-1} \hat{\mathbf{u}}_i) = \beta_i + \frac{1}{2} \mathbf{y}_i' (\mathbf{I}_N + \mathbf{X}_i \mathbf{V}_i \mathbf{X}_i')^{-1} \mathbf{y}_i. \end{aligned}$$

Using equation (2.4), one can easily obtain posterior samples or summaries of the entries of $\mathbf{U}$ and $\mathbf{D}$ conditional on $\boldsymbol{\theta}$. Instead of directly specifying the parameters of the NIG distribution, we use a vector of hyperparameters defined as $\boldsymbol{\theta} = (\theta_1, \theta_2, \theta_3)'$ based on the behavior of the Matérn covariance parameters. For the exponential covariance function with variance θ_1 and range $2/\theta_2$, assuming $m = 1$, parameters is computed as $d_i = \theta_1 (1 - \exp(-\theta_2 \|\mathbf{s}_i - \mathbf{s}_{J_1(i)}\|))$, $\alpha_i = 6$ and $\beta_i = 5d_i$, $\mathbf{V}_i = \text{diag}(v_{i1}, \dots, v_{im})$, $v_{ij} = \exp(-\theta_3 j)/d_i$. Regarding the choice of m , it is preferable to let m be data-driven, based on the decay of prior elements in the matrix $\mathbf{U}$. For computational simplicity, a uniform prior is commonly used: $P(\boldsymbol{\theta} \mid \mathbf{Y}) \propto p(\mathbf{Y} \mid \boldsymbol{\theta})$. Since the posterior distribution $P(\boldsymbol{\theta} \mid \mathbf{Y})$ is not analytically tractable, we use the adaptive Metropolis–Hastings algorithm to sample from it. Let $B = \{B_1, \dots, B_K\}$ be a partition of the index set $\{1, \dots, n\}$, define $U_k = \bigcup_{i \in B_k} J_i$, where J_i denotes the ordered neighbor set of location $\mathbf{s}_i$. Suppose index i belongs to block B_k . Define the new neighbor set $\bar{J}_i$ as $\bar{J}_i = \{j \in U_k : j \leq i\}$ and then apply Vacchiha approximation (Ghasemi et al., 2025). Simulation results indicate that posterior samples obtained from the grouping method leads fast computation and higher statistical accuracy of the estimates.

3 HSCL for Estimating Spatial-Temporal Covariances

Suppose the Gaussian distributed data $\{Z(\mathbf{s}_i, t), t = 1, \dots, T\}$ are given at irregular spatial locations $\mathbf{s}_i$, $i = 1, \dots, n$ and equally-spaced integer times $t = 1, \dots, T$ with a covariance function $C(\mathbf{h}, u)$ that can be parameterized by the vector $\boldsymbol{\theta} = (\theta_1, \dots, \theta_p)'$. Ignoring constants, $-2 [\log \text{ of the likelihood of } \boldsymbol{\theta} \text{ based on the observation vector } \mathbf{Z} = (Z(\mathbf{s}_1, 1) \dots, Z(\mathbf{s}_n, T))']$, is given by

$$L(\boldsymbol{\theta}) = \mathbf{Z}' \Sigma^{-1} \mathbf{Z} + \log |\Sigma|, \quad (3.1)$$

where Σ is the $nT \times nT$ covariance matrix of $\mathbf{Z}$. The full likelihood (3.1) involves $nT \times nT$ inversion matrix and hence is computationally difficult. Now suppose we consider all possible

contrasts of the type $Y_{ij}(t) = Z(\mathbf{s}_i, t) - Z(\mathbf{s}_j, t)$ for each $t = 1, 2, \dots, T$ and for all spatial sites $\mathbf{s}_i, \mathbf{s}_j$, which has a univariate stationary Gaussian process $\{Y_{ij}(t)\}$ with mean zero and covariance $C(\mathbf{h}, u)$ for fixed $\mathbf{h}$. Assume that the spectral density $2\gamma(\mathbf{h}, \cdot; \boldsymbol{\theta})$ exists. One possible version of CL can be based on the product of the marginal distribution of all possible pairwise contrasts $\{Y_{ij}(t)\}$. Assuming independence between the differences, ignoring constants, enables the negative composite log likelihood function can be written in the form

$$c\ell(\boldsymbol{\theta}) = \sum_{i=1}^{n-1} \sum_{j>i}^n (\mathbf{Y}'_{ij} \Sigma_{ij}^{-1} \mathbf{Y}_{ij} + \log |\Sigma_{ij}|), \quad (3.2)$$

where Σ_{ij} is the $T \times T$ covariance matrix of $\mathbf{Y}_{ij}$. The maximum CL estimate, $\hat{\boldsymbol{\theta}}$, is found by maximizing $c\ell(\boldsymbol{\theta})$ over the allowable parameter space. Notice that the CL (3.2) involves only T dimensional densities and hence is computationally simpler than the full likelihood (3.1). The computational burden can be further reduced if we apply the Whittle's idea.

$$c\ell(\boldsymbol{\theta}) = \sum_{i=1}^{n-1} \sum_{j>i}^n \sum_{k=0}^{T-1} \left(\frac{|J_{ij}(\tau_k)|^2}{2\gamma(\mathbf{h}, \tau_k; \boldsymbol{\theta})} + \log \gamma(\mathbf{h}, \tau_k; \boldsymbol{\theta}) \right), \quad (3.3)$$

where $|J_{ij}(\tau_k)|^2$ and $2\gamma(\mathbf{h}, \tau_k; \boldsymbol{\theta})$ are, respectively, the periodogram and the spectral density function defined at frequencies $\tau_k, k = 0, \dots, T-1$. HSCL estimates can then be obtained by minimizing (3.3) with respect to $\boldsymbol{\theta}$. Numerical optimization of $c\ell(\boldsymbol{\theta})$ is substantially simpler than the likelihood in the observation domain since the quadratic form, $\mathbf{Y}'_{ij} \Sigma_{ij}^{-1} \mathbf{Y}_{ij}$, and the determinant, $|\Sigma_{ij}|$, are not explicitly calculated. Also the parameters come in only through the spectral density, which for many types of classical time series has an explicit formula. The periodogram must be calculated, but it can be calculated quite easily via the fast Fourier transform (Mosammam, 2016, 2025).

4 Simulation and Real Data Analysis

To evaluate the performance of the Vecchia grouping methods, we first simulated 20 independent replicates from a Gaussian process with a Matérn covariance function with $\boldsymbol{\theta} = (3, 1, .25)$, $\mathbf{s}_i \sim U(0, 1)$ and simulations were conducted at 900 locations. To sample from the posterior distribution of the hyperparameters $\boldsymbol{\theta}$, we used the adaptive Metropolis–Hastings algorithm. Three Markov chains were run, each generating 30,000 posterior samples. After discarding the first 15,000 samples as burn-in, the remaining samples were thinned by a factor of 50, resulting in 900 independent posterior samples. Simulation results indicate that posterior samples obtained from the proposed method produce narrower uncertainty intervals compared to ungrouped approaches, suggesting higher statistical accuracy of the estimates (Figuer 1).

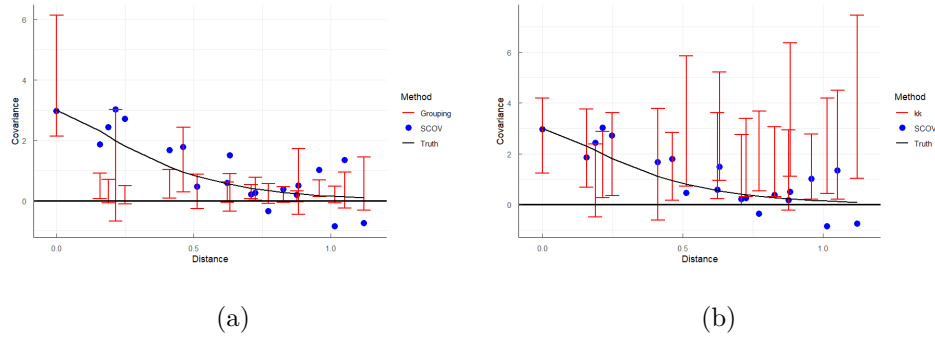


Figure 1: (a) Grouping method (b) Kidd and Katzfuss method.

Table 1: Fits of the ST Gaussian process with covariance model (4.1) to Irish winds data

Method	σ^2	α	a	c	β
WL	0.296(0.003)	0.956(0.002)	522.520(16.963)	1.270(0.042)	0.147(0.089)
HSCL	0.306(0.010)	1.003(0.002)	522.520(17.886)	1.151(0.088)	0.000(0.094)

Next we consider daily wind speeds data collected over 18 years, from 1961 to 1978, at 11 synoptic meteorological stations in Ireland. We fit the HSCL to the ST covariance function of

$$C(\mathbf{h}, u) = \frac{\sigma^2}{(|\mathbf{h}|/a)^\alpha + 1} \exp \left\{ -\frac{|u|/c}{((|\mathbf{h}|/a)^\alpha + 1)^{\beta/2}} \right\}, \quad (4.1)$$

where $a, c > 0$ are space and time scaling parameters, $\beta \in [0, 1]$ is the separability parameter and $\alpha \in (0, 2]$ governs the smoothness of purely spatial covariance function (Gneiting, 2002). The estimated parameters are summarized in Table 1 and details will be presented in presentation.

Discussion and Conclusion

First, we address a Bayesian grouping-based Vecchia's approximation approach which significantly reduces computational burden in nonparametric estimation of a nonstationary spatial covariance. Next, we propose HSCL for the estimation of ST covariances which requires no matrix inversions and the estimators are shown to be consistent and asymptotically normal under increasing domain asymptotic. Examples are given using simulated and real data.

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